L12.2: The Hessian matrix and local extrema for f(x,y)

Transcript Language

English

Hello and welcome to the match 2 component of the online Bsc program on Data Science and

Programming. In this video we are going to talk about the Hessian matrix and local extrema. So,

we have studied the notion of critical points for f x, y and in previous video we also studied the

Hessian matrix which consisted of the second order partial derivatives

placed in a square matrix. So, in the context of f x, y this will be a 2 by 2 matrix.

Let us recall first what was the second derivative test for

a one variable function. So, suppose you have a function of one variable defined in the domain D,

a point a in D is a critical point if either the function f is not differentiable at that point or

f prime is 0. And now if the function is twice differentiable, we applied something called the

second derivative test, so the second derivative test told us the nature of the critical points.

So, the test said the following; if a is a critical point and if the double derivative is

positive, then this point is a local minimum, if it is a critical point and the double derivative

is negative, then it is a local maximum and if it is a critical point and the double prime is 0,

then the test is inconclusive, so we do not know what happens in that case.

Fine, so this was the second derivative test for functions of one variable. Now, we would like to

have something analogous for higher order meaning functions, multivariable functions

and indeed the Hessian matrix is what is going to allow us to do that.

So, let us first recall critical points for multi-variable functions. So, if we

have a scalar valued multivariable function defined in our domain D in Rn, then a point

a in D is called a critical point if either the gradient is 0 or the gradient does not exist. So,

the main point here was that every local extremum, meaning a local maximum or a local minimum

was a critical point, so every local extremum is a critical point.

And what we want to do is once we get the collection of critical points

we want to identify which of these are local maxima, which of these are local minima

and if possible if which of these are saddle points. So, we know that not all critical points

are local extrema because there are things called saddle points and the saddle point is exactly a

critical point such that the gradient exists and the gradient is 0 but a is not a local extrema.

So, let us now see how we can use the hessian in order to classify these critical points.

So, again if f x, y is a function defined in the domain D and a is a critical point such that the

first and second order partial derivatives are continuous in an open ball around a. So,

this is a very important hypothesis, in particular this hypothesis allows us to

apply Clairaut s theorem and say that the mixed partials are going to be equal.

So, then the hessian test can be applied to check the nature of the critical point a

and what is the test? If the determinant of the Hessian matrix at that point is positive

and the second order partial with respect to x

at that point is positive, then it is a local minimum, if the determinant is positive and

the second order partial with respect to x is negative, then it is a local maximum and

if the determinant is negative, then it is a saddle point, and if the determinant is 0

then we do not know, the test is inconclusive. So, this is a slightly more involved test as you

can see then the second derivative test for one variable functions because here you have to

write down the matrix, compute its determinant and also check for an entry. Let us so, let us

try to understand how to keep these in mind. Suppose, I forget the Hessian test,

how do I remember what happens? Well, let us look at these functions,

so these functions we already know what the situation is, so you have x squared plus y

squared, we know how this function looks, it is like a bowl and 0, 0 is a global minimum, this is

something you just know from the fact that squares are positive, so 0, 0 is a local minimum, is a

global minimum in fact and so the test hopefully should tell us that let us see if that happens.

Similarly, minus x squared minus y squared is the same thing inverted down. So, now 0, 0 is a local

maximum in fact a global maximum but in particular global, local maximum. Similarly, x square minus y

squared 0, 0 was a saddle point, this was the prototype of how a saddle point look like.

And then x to the power 4 plus y to the power 4 here the, again 0, 0 is a local in fact a

global minimum, but since it is the fourth power something strange happens, so we. So, these are

the four prototype examples to keep in mind and these, so each of these will correspond to one of

the four conditions that we had before. So, first let us compute the gradient. So,

gradient of f here is 2x comma 2y, and from here we can compute the Hessian,

so the Hessian is 2 0 0 2, very nice Hessian. So, from here we get that the only critical point is,

so if I set my gradient to 0, then the only critical point is 0, 0 and the Hessian is

actually independent of the point, so the Hessian matrix for 0, 0 is in particular

2 0 0 2. And now let us apply the test. So, let us compute determinant of this matrix.

So, the determinant of this matrix is 4, so it is positive,

so good, so we know already that we are not in the inconclusive case and we know

that we do not have a saddle point because for saddle point the determinant will be negative.

So, now it will depend on the value of the 1, 1 term. So, here the 1, 1 term which is fxx

is 2, this is positive so therefore, it is a meaning 0, 0 is a local

minimum and this is exactly what we knew already by inspection because x square

plus y squared is always a positive function with its only 0 at 0, 0.

We can do the same thing for minus x squared minus y squared so here the gradient is minus

2x comma minus 2y. If I set it to 0 that gives me that the only critical point is

0, 0 and the Hessian here that means you take the partial derivatives of you take minus 2x

and take its partial derivatives so that is minus 2 and 0 and then you take minus 2y and take its

partial derivatives which is 0 and minus 2. So, notice what I did here by definition actually

I should have taken partial of this gradient with respect to x so that is minus partial of minus

2x and then partial of minus 2y, but I just took partials of minus 2x in the first row and partials

of minus 2 in the second and the reason that worked is because I know it is a symmetric matrix

from Clairaut s theorem. So, again this Hessian is

independent of whatever choice, so in (partic), meaning of whatever point, so in particular at

0, 0 you get the same Hessian and now let us see what happens so again the determinant is

non-zero, it is 4, so it is greater than 0. So, what that tells us is

one it is non-zero that means it is not in the conclusive situation and two that it is

positive which means it is either a local maximum or a local minimum

and that will be determined by whatever happens to fxx and fxx means the 1, 1th entry.

So, at 0, 0 this is minus 2 which is negative so therefore 0, 0 is a local maximum and this

indeed conforms to what we already know that this is a global maximum in fact.

Let us do x squared minus y squared, again if you compute the gradient,

this is 2x comma minus 2y, if you set it to 0 you get that the critical point is

x is 0 and y is 0, so that is 0 comma 0, let us find the Hessian, so for the Hessian you

differentiate 2x with respect to x and with respect to y, so with respect to

x it is 2, with respect to y it is 0, you take minus 2y differentiate with respect to x and y

with respect to x it is 0, with respect to y it is minus 2, and now you can see

it is a bit different than earlier. Of course, again it is independent of

the point so at 0, 0 also you have the same Hessian. And now what is the determinant,

so this determinant is 2 times minus 2 minus 0, so that is minus 4 which is negative

and you stop right here because once the determinant of the Hessian is negative in the

2 by 2 situation, then we know that this is a saddle point and indeed this is something that we

have seen already, so this is a saddle point. And finally let us do the example of x to the

power 4 plus y to the power 4, find the gradient 4x cube comma 4y cube, set it to 0 that means x

cube is 0, y cube is 0, that means x is 0, y is 0 so the only critical point is 0, 0.

Find the Hessian. So, now the Hessian actually involves functions,

so it is 12x squared 0 and then 0, 12y squared but what happens at 0, 0? So, at 0, 0 the Hessian

matrix actually is 0 0 0 0 and in particular that means that the determinant of the Hessian is 0

and therefore, this we cannot conclude anything from the test.

Therefore, the test is inconclusive and this is a warning that even for simple

functions like x to the power 4 plus y to the power 4 where we actually know that 0, 0 is a

global minimum not just a local minimum it is actually a global minimum because it is

a bowl which is x square plus y square is more like this, x to the power 4 plus y to the power

4 is more like this, so it goes up faster, 0, 0 is a global minimum.

But the test is inconclusive. So, these are the 4 prototypes of

corresponding to each of the cases that you can remember and if you remember these you will know

instead of by hearting the cases from here you can recall which case tells you what.

Let us do a couple of other examples.

So, this is x squared plus 6xy plus 4y squared plus 2x minus 4y. So,

let us find the gradient here, these are slightly more involved. So, 2x plus 6y plus 2 comma

6y, 6x plus 8y minus 4. So, if you set the gradient to 0, so equating to 0,

we get the critical point, we have actually done this before so I will not spend time here on this,

so we get the critical point 2 comma minus 1,

so this is something that we did in the video on critical points.

So, I will not repeat this computation, you get a system of 2 linear equations

in x and y and which you can solve, so we used Gaussian elimination to solve that, fine.

So, now let us ask what is the nature of this critical point, is it a saddle point, is it a

local minimum, is it a local maximum, can we say that from the test? So, for that we have to find

the Hessian matrix first. So, what is the Hessian matrix? So, you have 2x plus 6y plus 2 let us

take the partial with respect to x and y, so with respect to x you get 2, with respect to y you

get 6, so 6x plus 8y minus 4 the partial with respect to x is 6, with respect to y is 8.

And again this hessian does not depend on any point, so at 2 comma minus 1 also it is the same

matrix. Let us compute its determinant, so determinant of H f of 2 comma minus 1 is

8 times 2 16 minus 6 times 6 is 36,

so 16 minus 36 is minus 20 which is negative. So, this tells us that 2 comma 1 is a saddle point

of f x, y. So, let us view this function in Geogebra

and check exactly what is happening at 2 comma minus 1. So, here is the function f x, y is

x squared plus 6xy plus

4y squared plus 2x minus 4y. So, you can see that there is indeed

something interesting going on here, it looks like a, function looks quite nice actually.

And if we view it like this let us see what is happening at the point 2 comma minus 1. So,

here is the point 2 comma minus 1 and you can see right above the point 2 comma minus 1,

there is that saddle, so that saddle is coming at the point 2 comma minus 1 and indeed we are

being able to detect that using the Hessian matrix.

Let us do this second example which is f x, y is xy minus x cube minus y squared.

So, let us find the gradient so the gradient here is y minus 3x squared comma

x minus 2y. So, if we equate this to 0.

So, now we have 2 equations

x is 2y and y is 3x squared. So, let us substitute x is 2y in the first equation,

so if we do that. So, one ok so one point we already know at 0, 0, so the list of critical

points one point we already know, one is 0, 0 let us see if there is any other point.

So, for that we substitute x is 2y in the first equation, so we get y is equal to 6 times

2y, my bad, 3 times 2y squared which is 12 times y squared which means y times

1 minus 12 y is 0. So, one option is y is 0 that is exactly the point 0 comma 0 that

corresponds to the point 0 comma 0. The other option is that y is not 0

in which case 1 minus 12y must be 0, so which means y must be 1 by 12. And if y is 1 by 12,

then we can get x is 2 by 12 so which is 1 by 6 so this 1 by 6 comma 1 by 12.

So, there are two critical points 0, 0 and 1 by 6 comma 1 by 12. Let us check the nature of these

critical points, so for that we have to find the Hessian matrix. So, I will take partials again,

so y minus 3x squared the partial with respect to x is minus 6x with respect to y is 1, x minus 2y

partial with respect to x is 1 and partial with respect to y is minus 2. So, at the point 0, 0

this becomes 0 1 1 and minus 2, so the determinant is negative, so this is minus 1 which is less than

0 and that tells us it is a saddle point. So, therefore 0, 0 is a saddle point.

The other critical point is 1 by 6 comma 1 by 12, let us find out what the Hessian is.

So, if I substitute 1 by 6 in the 1, oneth

place, I get minus 1 so minus 1 1 1 minus 2 so the determinant of the Hessian matrix

is 2 minus 1 which is 1, so this is positive, so we know this is either

a local minimum or a local maximum. And for that it is determined by the 1,1th term, so f x, x at

of 1 by 6 comma 1 by 12 so that is a 1, oneth term which is this term here and that is minus 1

which is less than 0, so therefore 1 by 6 comma 1 by 12 is a local maximum.

Let us see in Geogebra if this is indeed what is happening.

So, this is how the graph of the function looks like in Geogebra and you can see at 0,

0 it really has very interesting behavior

and that it is clearly a saddle point. So, if we zoom in, let us see if I can come close to 0, 0.

So at 0, 0 you can see at if you take the, take it along the green line,

then the function values it is like this, so it is a local maximum, whereas if you take it along the

other axis it is decreasing like this, increasing a little and then

it again goes down, so it is a local minimum along that, so indeed it is like a saddle point,

although it looks a bit different than the standard saddle which is a little more

clear to see. But it is clear that at 0, 0 something strange happens.

The more interesting point is 1 by 6 comma 1 by 12 that is a bit hard to pick up from the

graph. So, here is that point, it is a bit difficult to see but let us zoom in further and

maybe we can see what is there, so now we can see this point here, here is the point

b which is 1 by 6 comma 1 by 12 and you can see that close to this point the function is indeed

sort of positive, so it is coming down, remember at before 0, 0 from the

on the other side it was coming down but then just for a brief period it kind of goes up again

and then it goes down and that is what is being picked up by our Hessian matrix over there.

So, this indeed conforms to our understanding of what is happening. So, maybe here one can see,

yeah. So, you can see from the graph it may not be very easy to pick up these points but the algebra

tells you what is happening and even from the function directly the equation of the function

it is not clear why this is a local maximum but the second order test is going to,

the Hessian test is going to tell you that indeed it is a local maximum.

Let us do this last example of f of x, y sin of xy. So, we have computed the gradient for sin

of xy, which let us write that down again, this was y times cosine xy comma x times cosine xy.

In fact we also wrote down the Hessian, the Hessian was minus y squared sin of xy,

then the other diagonal term is minus x squared sin of xy and then the off diagonal terms were

slightly more complicated. So, cosine of xy minus xy times

sin of xy and then the same thing over here by Clairaut theorem or by computation.

So, this is how our Hessian matrix looks like, so first what are the critical points.

So, we actually know for this function what is happening, we have seen a

graph of this function and several times and we know what is happening for this function because

the local maxima or minima are actually global maxima or minima and they are given by

when the function this is a sin function, so it takes the value 1 and minus 1 respectively.

So, when does it take the value 1 when x times y is of the form pi by 2 plus

k times 2 pi where k is an integer. So, for all such points the function takes the value 1, so the

value 1 will be taken whenever x times y attains one of those values so it is curves and we have

actually seen this picture, we will see it again. And similarly, when does it take the value minus 1

when it is of the form 3 pi by 2 plus k times 2 pi where k is an integer again

and again that is a curve and on those points it takes value minus 1.

And the only other point which ends up being a critical point is the following. So,

let us equate, so equating gradient to 0, we get, so two things can happen, one is

cosine of xy is 0, the other is if cosine of xy is not 0 then both x and y should be 0.

So, cosine of xy is 0 is exactly the set of points that I just said, so cosine of xy 0 means that

sin of x y is plus or minus 1 because we know that sin squared plus cosine squared is 1.

So, this is the same as saying that sin of xy is plus or minus 1, so plus 1 is exactly the

set of local, global in fact maxima and minus 1 is exactly the set of global minimum. So, we

know actually what happens at all these points, so these are points that we are already understand.

What happens at x is equal to y is equal to 0, so this is the point that we do not know what

happens. Well, it is fairly clear that this is probably not a point of local maximum or local

minimum, so most likely this is a saddle point and let us see if our Hessian is telling us that.

So, if you, if we compute our Hessian what do we get? So, Hessian at 0, 0, so I should say Hessian

at 0, 0, well, substitute x is y is 0, so the diagonal entries become 0 what happens to the off

diagonal entries? Well, x, y times sin xy becomes 0, so the only contributing term is cosine of xy

that is cosine of 0 when x is y is 0 and cosine of 0 we know is 1, so this is the matrix we get,

this implies determinant of this matrix, so sometimes this is called the Hessian determinant

is minus 1 and that tells us that 0, 0 is a saddle point for this function.

So, the point that was sort of we were unsure of what happened,

the test told us it is a saddle point. Let us see what happens for those points

for which we are sure of what happens, let us see what happens to points where

of this form where cosine of xy is 0 or sin of xy is plus or minus 1. So, for points such that

cosine xy is 0 let us compute the Hessian.

So, as I said there are two options either sin of xy is 1 or sin of xy is minus 1, let us say sin of

x y is 1. So, there are two options so I will say this or

this and we will see what happens in both cases.

So, if sine of xy is 1 then this gives me minus y squared minus xy minus xy minus x square.

And if sine of xy is minus 1, then this gives me y squared x squared xy and xy.

So, I will just reiterate that this is if sine of xy is 1 and this is if sine of xy is

minus 1. And what happens to the determinant? So, suppose we compute the determinant so the

Hessian determinant. Well, you can see what is happening, this is x squared y squared

minus x squared y squared which is actually 0. So, this test is inconclusive even though we

actually know what is happening at these points and the same thing happens here,

this is also 0, x squared y squared minus x squared y square.

So, the point of this example is to say that the Hessian test is a, has several features to it and

what can happen is that points where it is obvious that there are local maxima or local

minima it may end up being inconclusive, whereas points that we actually do not know

what is happening from the function it gives us a result, so such things can happen.

So, the Hessian test it is not a full proof test but it is our first step towards understanding

critical points. So, we have studied the Hessian test in this video for f x,

y we will study this in the forthcoming video for more number of variables. Thank you.